

HSH HEAVY SAFE HELICOPTER

HSH-00 | REVISION 20

FINAL GENERATION 1 COMPREHENSIVE MASTER ENGINEERING BASELINE

Founder: Amirbahador Atai | Document: HSH-00 | Revision: Rev 20

Revision 20 is the definitive Generation 1 master engineering baseline. It incorporates every architectural correction established through Rev13–Rev19, adds four critical engineering enhancements (corrected seat specification, sealed-capsule airbag hazard analysis requirement, revised airbag mass budget, IP-D system-level claim), and adds five new technical sections (TRL table, competitive differentiation, defined-vs-open summary, preliminary capsule geometry, and mission profile). The result is a document suitable as the primary technical data-room instrument for IP licensing evaluation by an industrial acquirer. This document is not a certification approval, not a filed patent, and not proof of achieved performance.

DOCUMENT CONTROL

Field	Value
Document Title	HSH Heavy Safe Helicopter — Generation 1 Comprehensive Master Engineering Baseline
Document Number	HSH-00
Revision	Rev 20
Supersedes	HSH-00 Rev 12 through Rev 19
Status	FINAL G1 CONCEPTUAL MASTER BASELINE
Founder	Amirbahador Atai
Development Stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Classification	Proprietary Concept Development Material
Certification note	Not a certification approval
Patent note	Not a filed patent

Engineering Integrity Statement: TARGET, ASSUMPTION, STUDY, CALC and OPEN values are not demonstrated performance. Critical claims require analytical closure, simulation, ground testing, integrated testing and flight-test evidence. Every status label in this document shall be interpreted precisely as defined in §5.

REVISION HISTORY

Rev	Primary Change	Status
Rev 12	26,000 kg / 26.0 m / 4.0 m baseline; 8-canopy recovery; single 6 s target; 10–12 occupants	Superseded
Rev 13	Main-rotor separation removed from core; staged S0–S5; 3+2 canopies; Risk Register; Prior Art	Superseded
Rev 14–16	Full narrative depth; helium-hybrid trade posture; competitive IP narrowing	Superseded
Rev 17–18	HSH identity unified; 8-person lock; commercial content separated; Maturity Framework	Superseded
Rev 19	16-airbag architecture; seat specification; OI-16/OI-17 registered	Superseded
Rev 20	Seat spec corrected; sealed-capsule airbag OI-18; airbag mass 200 kg; IP-D; TRL; competitive diff; defined-vs-open; geometry; mission profile	CURRENT

TOP-LEVEL TECHNICAL BASELINE — MASTER TABLE

Parameter	Rev 20 Value	Status	Key Note
Design mass	26,000 kg	[TARGET]	Planning closure; bottom-up pending
Maximum capsule occupancy	8 persons	[LOCKED LIMIT]	Generation 1 hard limit
Main rotor diameter	26.0 m	[TARGET]	Benchmark comparison §4.1
Main rotor disk area	530.93 m ²	[CALC]	$\pi \times (13.0)^2$
Disk loading	48.97 kg/m ²	[CALC]	See §4.1 for benchmark
Tail rotor diameter	4.0 m	[TARGET]	Anti-torque closure pending
Tail rotor disk area	12.57 m ²	[CALC]	$\pi \times (2.0)^2$
Capsule mass	7,000 kg	[TARGET]	Bottom-up pending Doc 113
Range target	1,000 km	[TARGET]	Mission model pending §21
Ideal hover power	≈ 3.57 MW	[CALC]	Actuator disk, ISA SL
Shaft power @ FM 0.75	≈ 4.76 MW	[CALC]	Study value only
Propulsion	Two turboshaft engines	[REQ/STUDY]	Engine selection open
Overall length	16.5–18.0 m	[TARGET/STUDY]	Layout reconciliation pending
Fuselage width (max)	4.0–4.5 m	[TARGET/STUDY]	
Overall height	4.8–5.3 m	[TARGET/STUDY]	
Main rotor separation	Not part of core safety case	[DEFERRED — RESEARCH TRACK]	§12
Recovery canopies	3 primary + 2 reserve	[STUDY]	§14
Separation timeline	Staged S0–S5	[TARGET/OPEN per stage]	§13
Occupant airbags	16 total (2 per occupant)	[TARGET/STUDY]	§10A; sealed-capsule hazard OI-18
Crashworthy seat stroke	≥150 mm dynamic stroke	[TARGET/STUDY]	§10B; replaces Rev19 foam thickness
Peak spinal load limit	≤6,670 N reference criterion	[TARGET/STUDY]	FAR 27.562 proxy; §10B
Development gate	G0 approaching closure	[CURRENT]	G1–G8 not started

MASS ACCOUNTING — 26,000 kg

Mass Group	Planning Mass	Status
Survival capsule, complete	7,000 kg	TARGET
Main airframe excluding capsule	5,000 kg	ASSUMPTION / STUDY
Main rotor, hub and controls	2,100 kg	ASSUMPTION / STUDY
Tail rotor, gearbox and drive	480 kg	ASSUMPTION / STUDY
Engines, transmission, gearboxes and mounts	3,100 kg	ASSUMPTION / STUDY
Fuel system + full usable fuel	3,400 kg	ASSUMPTION / STUDY
Rapid separation system	1,000 kg	ASSUMPTION / STUDY
Avionics, electrical, hydraulic, thermal, control	1,600 kg	ASSUMPTION / STUDY
Operating fluids and consumables	320 kg	ASSUMPTION / STUDY
Growth / configuration reserve	2,000 kg	PROGRAM RESERVE
TOTAL	26,000 kg	CALC / ACCOUNTING CLOSURE

CAPSULE COMPONENT MASS — 7,000 kg

Capsule Component	Planning Mass	Notes
Primary crashworthy cell / internal roll-cage	1,950 kg	CFRP + titanium nodes
Outer capsule shell / fairing / thermal protection	500 kg	
Eight fixed landing supports + progressive energy absorbers	850 kg	
Eight crashworthy seats + restraints (≥150 mm stroke, ≤6,670 N)	700 kg	Rev 20 corrected spec
Sixteen occupant airbags + inflators + sensors + wiring	200 kg	Rev 20 corrected (was 80–120 kg)
Emergency exits / doors, frames and seals	250 kg	
Distributed multi-canopy recovery system (3 primary + 2 reserve)	650 kg	
Independent capsule emergency power + batteries	250 kg	
Life support / emergency oxygen / environmental	200 kg	
Avionics / sensing / communications / locator / lighting	250 kg	
Separation interfaces + capsule-side release hardware	350 kg	
Internal secondary structure / floor / partitions / padding	300 kg	Reduced to absorb airbag mass
Fire / smoke detection and suppression	150 kg	
Water landing / flotation / sealing	150 kg	
Inspection / access / replaceable protection	100 kg	

Capsule Component	Planning Mass	Notes
Capsule configuration growth allowance	150 kg	
TOTAL	7,000 kg	CALC / CLOSURE

***Rev 20 mass note:** Airbag system mass corrected from 80–120 kg (Rev 19) to **200 kg** based on automotive-class benchmarks scaled to 16-bag aviation-grade system with ECU, pyrotechnic harness, mounting and qualification margin. Corresponding reduction taken from internal secondary structure line. Total capsule budget unchanged at 7,000 kg. If airbag detailed design exceeds 200 kg, growth shall be absorbed by the 150 kg configuration growth allowance; any further growth triggers configuration-controlled impact assessment.*

1. BASELINE FREEZE AND CONTROLLED IDENTITY

1.1 Programme Identity

The programme name is **HSH Heavy Safe Helicopter**. The controlling document is **HSH-00 Revision 20**, Founder: Amirbahador Atai. All prior references to HSC-00 or any earlier HSH-00 revision are superseded. Rev 20 is the final Generation 1 conceptual master baseline.

1.2 Occupancy Lock

8 persons maximum [LOCKED LIMIT]. This is a Generation 1 hard configuration constraint. Any future change requires formal configuration control and re-analysis of geometry, mass, CG, inertia, seating, restraints, airbag count, life support, egress, structural loads, recovery sizing and certification basis.

1.3 What Rev 20 Changes Relative to Rev 19

Change	Rev 19	Rev 20	Rationale
Seat specification	≥50 cm ergonomic foam	≥150 mm dynamic stroke; ≤6,670 N spinal load	Foam thickness is not an aviation crashworthy-seat criterion
Airbag mass budget	80–120 kg	200 kg	Automotive benchmark scaled to 16 aviation-grade units
Sealed-capsule airbag hazard	Not explicitly registered	OI-18 added	16 airbags in sealed volume: overpressure and gas-toxicity risk
IP-D claim	Not present	Added	System-level combination claim stronger than any single IP-A/B/C
TRL table	Not present	Added §22	Required by any serious technical acquirer
Competitive differentiation	Not present	Added §23	Standard due-diligence document
Defined-vs-open summary	Not present	Added §3	First document a diligence team reads
Capsule geometry estimate	Not present	Added §24	Geometric plausibility of 8 occupants + airbags + stroke seats
Mission profile table	Not present	Added §21	Anchors the 1,000 km range requirement

1.4 What Rev 20 Does Not Change

Every requirement, status label, open item (OI-01 through OI-17), risk register entry, verification gate, interface definition, safety layer, Safety Kernel precondition set, staged separation timeline, recovery architecture, prior art posture, and engineering maturity framework from Rev 18 and Rev 19 is retained in full and unchanged in Rev 20, except where explicitly corrected in the change table above (seat criterion, airbag mass, OI-18, IP-D, and new sections).

2. PURPOSE AND SCOPE

2.1 Programme Objective

HSH Generation 1 defines the system-level architecture for a 26-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain — not reliance on any single device.

2.2 Core Scoping Decision

The survival chain is designed to work assuming the main rotor is never separated from the aircraft. This deliberate decision was introduced in Rev 13 and is retained in Rev 20. No credible trajectory, gyroscopic, or structural analysis yet exists to support main-rotor separation from a descending occupied capsule. Building the core safety case on an unproven capability would undermine confidence in the entire architecture and in all primary IP claims.

2.3 Scope Coverage

This document covers: the aircraft; the occupied survival capsule; the rapid selective separation system (non-rotor components only); safety initiation paths; deterministic safety authorization; stabilization; recovery; impact attenuation; post-event survival; and verification interfaces. It does not cover detailed structural analysis, certified engine data, qualification test results, airworthiness findings, or IP legal opinions — those are evidence-gate deliverables.

2.4 Design Philosophy

Separation is a controlled emergency mode, authorized only when defined conditions are satisfied. The survival architecture supports multiple outcomes in priority order: (1) conventional flight recovery; (2) autorotation; (3) selective hazard/mass reduction; (4) capsule separation and parachute recovery; (5) energy-absorbing landing; (6) post-landing rescue. The most extreme mode shall not be required for manageable failures.

2.5 Human Authority

Human emergency authority is preserved and non-bypassable. No AI system, optimization algorithm, or predictive model shall override the crew's ability to command, inhibit, or terminate an automatic survival sequence. This is a design requirement, not a guideline.

3. WHAT IS DEFINED TODAY vs WHAT IS OPEN

This section is new in Rev 20. It is the first thing a technical due-diligence team reads.

Domain	What is Defined (Rev 20)	What Remains Open	Gate
System architecture	Six-layer survival; Safety Kernel; state machine; staged separation; 3+2 recovery; 8-support landing	All engineering analysis	G1

Domain	What is Defined (Rev 20)	What Remains Open	Gate
Mass accounting	26,000 kg and 7,000 kg planning models; component-level allocation	Bottom-up verification; uncertainty and growth	G1
Capsule structure	CFRP roll-cage concept; titanium nodes; load path description	FEA; crash load analysis; fatigue	G5
Separation system	S0–S5 timeline; candidate non-rotor components; mandatory protections	S3 mechanism; 6-DOF; ground demonstration	G2–G3
Recovery system	3+2 canopy architecture; sizing rationale; failure tolerance concept	Canopy sizing; inflation dynamics; min deployment altitude	G6
Landing supports	8-point concept; energy absorber; asymmetric-failure requirement	FEA; drop tests; terrain interaction	G5
Crashworthy seats	≥150 mm stroke; ≤6,670 N criterion; 8 positions	Seat design; foam/stroking-leg; dynamic tests	G2–G5
Airbag system	16-bag concept; 2 per occupant; placement; 200 kg budget	Packaging; deployment logic; sealed-volume gas (OI-18); sled tests	G2–G5
Safety Kernel	Precondition set defined; verification requirements stated	Software/hardware implementation; formal verification	G1–G2
IP claims	IP-A, IP-B, IP-C, IP-D conceptually defined; prior art posture stated	Professional prior-art search; claim drafting; filing	External legal
Certification basis	CS-29 referenced	Formal regulatory engagement; means of compliance	External regulatory
Performance	Hover power study (3.57 MW); disk loading benchmark	All performance analysis; cruise; OEI; range closure	G1
Mission profile	1,000 km range target; 8 occupants	Fuel model; cruise speed; altitude; reserve policy	G1

4. AIRCRAFT GEOMETRY AND CONFIGURATION

4.1 Rotor Disk Loading — Benchmark Comparison

Aircraft	MTOW	Main Rotor Diameter	Disk Loading
Mi-26 (Russia — heaviest production)	56,000 kg	32.0 m	~55 kg/m ²
CH-53K King Stallion (US heavy-lift)	38,400 kg	24.1 m	~58–60 kg/m ²
CH-47F Chinook (tandem, per disk)	22,680 kg	18.3 m per rotor	~43 kg/m ² per disk
Sikorsky S-92 (medium offshore)	12,020 kg	17.17 m	~52 kg/m ²
AW139 (medium — market neighbor)	6,800 kg	13.8 m	~45 kg/m ²
HSH Gen1 Rev 20	26,000 kg	26.0 m	48.97 kg/m ²

Design rationale: HSH's disk loading is deliberately lower than Mi-26 and CH-53K. This trades rotor size for: (a) greater hover power margin — needed given the additional mass of the capsule, separation system, airbags and crashworthy seats; (b) reduced downwash velocity at the eight fixed landing supports; and (c) lower tip-speed requirement, reducing rotor noise. Confirmed or revised at Document 063 closure.

4.2 Aircraft Geometry Summary

Geometry Item	Preliminary Definition	Status
Overall length	Approx. 16.5–18.0 m	[TARGET/STUDY]
Maximum fuselage width	Approx. 4.0–4.5 m	[TARGET/STUDY]
Overall height	Approx. 4.8–5.3 m	[TARGET/STUDY]
Main rotor diameter	26.0 m	[TARGET]
Tail rotor diameter	4.0 m	[TARGET]
Rotor arrangement	Single main rotor + tail rotor	[REQ/STUDY]
Landing supports	8 fixed, short, mechanical supports under capsule	[TARGET]
Retractable landing gear	None	[LOCKED CONCEPT]
Capsule position	Central, integrated within primary airframe	[LOCKED CONCEPT]
Capsule normal-flight load path	Primary attachment system with controlled emergency release	[REQ]
Capsule emergency load path	Independent final landing element after verified clearance	[REQ]

5. BASELINE STATUS AND CONFIGURATION CONTROL

5.1 Status Label Definitions

Status Label	Precise Meaning
[LOCKED CONCEPT]	Retained as baseline design principle; not to be changed without formal impact assessment
[LOCKED LIMIT]	Hard Generation 1 constraint; requires formal change process and full re-analysis to alter
[REQ]	Top-level requirement; binding on the design unless formally waived
[TARGET]	Numerical or functional development objective; not demonstrated
[ASSUMPTION / STUDY]	Working engineering assumption for preliminary sizing; to be replaced by analysis or test
[CALC]	Calculated from stated assumptions; inputs are themselves assumptions or targets
[OPEN]	Requires analysis, simulation, test, supplier data or design decision
[OPEN — CRITICAL]	Open item with material effect on the core safety case; priority closure required
[DEFERRED — RESEARCH TRACK]	Excluded from licensable core baseline; not to be presented as core product
[TBD — DEPENDENCY]	Cannot be fixed until a dependent activity closes
[PRIOR ART STATUS: NOT YET SEARCHED]	No novelty conclusion implied; professional search required

5.2 Configuration Change Triggers

A formal configuration change is required for any of the following: MTOW; main rotor diameter; tail rotor diameter; capsule mass; occupant count; separation sequence architecture; parachute architecture; landing-support count or geometry; airbag count or placement; seat stroke specification; Safety Kernel precondition set; any safety-layer definition. Each change triggers an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

6. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

The six layers are **functionally distinct** and **independently capable** — not merely redundant. Independence means physically, electrically, and software-separate implementations that cannot both be defeated by any single common-cause failure.

Layer	Function	Primary Purpose	Independence Requirement
L1	Primary automatic emergency detection and initiation	Detect catastrophic states; initiate validated survival sequence when Safety Kernel authorizes	Separate processor, sensor suite, power bus from L2
L2	Independent backup initiation	Second segregated automatic path if L1 unavailable or in disagreement	Different physical location, software architecture, power source from L1
L3	Independent manual/crew authority	Crew initiation, inhibit and emergency authority without L1/L2	Hardwired path bypassing L1/L2; software cannot disable
L4	Rapid selective physical separation	Release designated non-rotor hazards using controlled mechanisms	Dual independent actuation paths; no single-point failure defeats both

Layer	Function	Primary Purpose	Independence Requirement
L5	Direct-impact survival	Protect occupants via capsule structure, restraints, seats and eight energy-absorbing supports	Passive structural system; no active components required for primary function
L6	Recovery and post-event survival	Deploy recovery canopies; locate occupants; support rescue	Independent emergency power; independent comms; independent locator beacon

AI Boundary — Non-Negotiable Principle: AI or advanced prediction may assist detection, classification and decision support. **Deterministic safety logic authorizes safety-critical actuation. AI confidence is never equivalent to safety authorization.** This principle is non-negotiable from a certification perspective and is the engineering basis for IP-A (§17).

Survival Chain: PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE (NON-ROTOR) → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT (8 supports + crashworthy seats + 16 airbags) → SURVIVE / LOCATE / RESCUE → AUDIT

7. DETERMINISTIC SAFETY AUTHORIZATION AND SAFETY KERNEL

The Safety Kernel prevents AI, digital-twin models, and predictive systems from directly commanding hazardous separation without satisfying a defined set of physically verifiable preconditions. It is the boundary between the decision layer and the actuation layer.

7.2 Safety Kernel Precondition Set

Precondition Category	Required Evaluation
Altitude and airspeed	Above minimum recovery altitude; within separation dynamic-pressure envelope; within parachute deployment envelope
Attitude and angular rates	Pitch, roll, yaw within acceptable separation envelope; angular rates within acceptable limits
Rotor and engine state	Rotor RPM documented; engine/fire/fuel hazard state known and consistent with separation sequence
Structural and attachment	Capsule attachment health confirms release mechanisms functional; structural health consistent with planned separation
Trajectory and clearance	Acceptable clearance between released components and capsule/parachutes confirmed
Recovery system	Parachute deployment envelope satisfiable after separation; recovery integrity confirmed
Power and communication	Independent survival power available; communication loss does not inhibit Safety Kernel
Human authority	Manual crew inhibit confirmed; rollback/fallback availability assessed
Airbag authorization	Impact state valid; deployment permitted under deterministic preconditions; capsule-interior environment suitable

7.3 Kernel Decision Table

Decision	Minimum Design Question	Verification Method
Authorize separation	Are all mandatory preconditions simultaneously true?	Formal logic review + fault injection
Inhibit separation	Can an unsafe state block actuation even under software fault?	Fault injection + power-loss test
Fallback mode	What happens if a sensor or command path is lost?	Failure-mode testing per FMEA
Human override	Is authority bounded, deterministic and physically independent?	HMI + system integration test
Airbag authorize	Is impact state valid and deployment safe in closed capsule?	Logic review + sled/impact test + gas analysis
Audit record	Can every emergency event be fully reconstructed?	Data-recording verification test

The Safety Kernel shall be implemented with: formally verified software or hardware logic; documented requirement set with code traceability; independence from AI/ML layers; segregated power supply; independent communication path to actuation hardware; protection against command injection; and an immutable audit record. No update or in-flight reprogramming shall modify the Safety Kernel precondition set without a formal change process and re-verification.

8. CONTROLLED RAPID SEPARATION SYSTEM

The system is selective, staged, and controlled — not a blanket jettison of everything. State-dependent release logic means the set of components released in a fire emergency differs from the set in a structural emergency. Every component designated as separable must be confirmed through structural, aerodynamic, dynamic, fire, control, trajectory and collision analysis.

8.2 Candidate Separable Components

Component	Condition for Separation	Analysis Requirement
Main engines (both)	After fuel isolation; where structurally separable	Trajectory, structural, thermal, fire analysis
Selected transmission/drive components	Where release is structurally and dynamically acceptable	6-DOF dynamics, CG shift analysis
Tail rotor assembly + gearbox + drive	Where aerodynamically and dynamically safe	Anti-torque state, trajectory, collision analysis
Primary fuel tanks + major fuel lines	After confirmed fuel isolation	Fire hazard analysis, isolation timing
Selected external structural panels	High-mass, not on capsule primary load path	Structural, trajectory, collision analysis
Main rotor blades and hub	EXCLUDED from core candidate list	Deferred research track — §12.4

8.3 Mandatory Release Protections

Protection	Requirement
Positive mechanical retention	Active positive lock during all normal flight; release requires deliberate actuation
Electrical/logic inhibit	Release electrically inhibited in normal flight; specific arming sequence required
Independent command paths	Primary and backup paths physically and electrically independent
Defined arm/fire sequence	Documented, tested, protected against out-of-sequence commands
Release-state sensing	Positive confirmation of attached/released state at each stage
Fuel/energy isolation	Isolation precedes or is concurrent with release; no new uncontrolled hazard
Asymmetric/partial-release detection	Full reporting of any case where less than the designated set is released
Post-release monitoring	Track released components where practically possible
Capsule-clear verification	Capsule release not authorized until all non-rotor components confirmed clear
Failed-release fallback	Safe state defined for every possible failed-release configuration

8.4 Separation System Mass Budget — 1,000 kg

Separation System Component	Mass
Primary capsule release mechanisms and locks	240 kg

Separation System Component	Mass
Independent backup release mechanisms	150 kg
Local structural reinforcement and release fittings	160 kg
Independent initiators / actuators	90 kg
Release-state sensors and monitoring	65 kg
Independent power / signalling and protected wiring	70 kg
Fuel / energy isolation interfaces	55 kg
Retention, inhibit and safety-interlock hardware	50 kg
Configuration growth allowance + integration margin	120 kg
TOTAL	1,000 kg

9. OCCUPIED SURVIVAL CAPSULE

The capsule is not a cabin with a parachute. It is a structurally independent survival system that functions as the passenger compartment during normal flight and becomes the final landing boundary in a catastrophic emergency. Every subsystem inside the capsule must function independently of the main airframe after separation. Primary structure: continuous CFRP roll-cage with metallic/titanium nodes at load-introduction points.

Capsule Parameter	Rev 20 Definition	Status
Occupancy	8 persons maximum	[LOCKED LIMIT]
Complete capsule mass	7,000 kg planning target	[TARGET]
Primary structure	Continuous CFRP roll-cage, titanium nodes	[CONCEPT]
Seat type	Energy-absorbing, ≥150 mm stroke, ≤6,670 N peak spinal load	[TARGET/STUDY]
Airbag system	16 total (2 per occupant)	[TARGET/STUDY]
Landing supports	8 fixed, progressive energy absorbers	[TARGET]
Recovery canopy architecture	3 primary + 2 reserve	[STUDY]
Independent emergency power	Battery-based, independent of airframe bus	[TARGET]
Life support	Emergency oxygen, minimum post-landing endurance	[TARGET]
Capsule-clear verification	Required before recovery deployment	[REQ]
Post-landing stability	8-point architecture; asymmetric/failed-leg analysis required	[REQ]
Water landing	Flotation, sealing for ditching scenario	[TARGET]
Emergency egress	Multiple exits rated for crash loads; airbag-compatible paths	[TARGET]

10. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

Eight short, thick, fixed mechanical supports are integrated with the capsule lower structure. They are not conventional helicopter landing gear. Capsule-integrated: supports remain with occupants after separation. Redundancy: loss of one or two supports still leaves six or seven functional. Energy absorption: each support incorporates a progressive mechanical energy-absorber with controlled, bounded crush force throughout deformation without a hard metallic stop.

Impact Velocity	Total KE (7,000 kg capsule)	KE per Support (÷8)
6 m/s	126.0 kJ	15.75 kJ
8 m/s	224.0 kJ	28.0 kJ
10 m/s	350.0 kJ	43.75 kJ

These are sizing illustrations, not allowable impact energies. Seat stroke and airbags provide additional occupant load management after the supports arrest capsule motion.

Case	Analysis Focus	Required Evidence
All 8 supports, flat terrain	Load sharing and stability	FEA + drop test
1 support failed	Redistribution and local overload	FEA + test
2–3 supports failed	Survivability boundary	Sensitivity study
Sloped terrain (≤15°)	Roll/tip tendency, asymmetric load transfer	Terrain simulation
Soft ground / snow	Penetration and energy dissipation	Ground interaction test
Water impact (ditching)	Hull sealing and flotation	Water impact test
Oblique combined loads	Lateral + vertical + longitudinal	6-DOF dynamic simulation

10A. SIXTEEN-AIRBAG OCCUPANT PROTECTION ARCHITECTURE

Item	Rev 20 Baseline
Occupant positions	8 (locked maximum)
Airbags per occupant	2
Total airbags	16
Primary set	8 — one per seat, head/torso / principal impact direction
Secondary set	8 — one per seat, lateral / lower-body / secondary direction
Status	[TARGET/STUDY]
Mass planning (Rev 20 corrected)	≈ 200 kg (16 aviation-grade modules + inflators + ECU + harness + mounting)
Verification	Sled tests, inflation tests, interference checks, gas-toxicity review, egress compatibility

Placement principles: Eight primary + eight secondary; shall not obstruct egress; compatible with CFRP roll-cage, seats and restraints; deployment only under deterministic Safety Kernel preconditions; airbag authorization passes through Safety Kernel — not an independent system that can deploy outside the authorization chain.

Rev 20 mass correction: Rev 19 estimated 80–120 kg. Aviation-grade modules at 1.5–2.0 kg each × 16 ≈ 24–32 kg module mass. Adding ECU (8–15 kg), pyrotechnic harness (20–30 kg), mounting (30–40 kg), wiring (15–20 kg), and qualification margin yields approximately 150–220 kg, rounded to **200 kg** planning value. The Rev 19 estimate was insufficient and would have created a false closure of the capsule mass budget.

Sealed-Capsule Deployment Hazard — Rev 20 Critical Addition (OI-18): Sixteen airbags deploying simultaneously in a sealed capsule create risks not present in automotive applications: (a) overpressure — rapid gas generation may transiently exceed safe ear-drum and sinus pressure limits; (b) gas composition — propellant gases dilute capsule oxygen; (c) temperature — inflator heat in a sealed volume; (d)

particulate — sodium compounds from certain propellant chemistries. Registered as **OI-18 (Critical)**. Mitigation options include staggered sequential deployment, vented inflators, or reduced inflator energy for the closed-capsule environment.

10B. EIGHT CRASHWORTHY SEATS — REV 20 CORRECTED SPECIFICATION

Rev 19 specified “≥50 cm ergonomic foam seating-surface thickness.” This is not a recognized aviation crashworthy-seat criterion. FAR 27.562 (and CS-29 equivalent) evaluates seats by occupant injury criteria, not foam thickness. Soft foam can compress fully under crash loads in milliseconds, providing minimal protection after initial compliance. The correct engineering specification is a minimum dynamic stroke and a peak load limit.

Item	Rev 20 Baseline
Number of seats	8
Minimum dynamic stroke	≥ 150 mm (combined seat/absorber/floor deformation path)
Peak spinal compressive load limit	≤ 6,670 N (FAR 27.562 reference criterion, proxy)
Force-displacement response	Progressive — bounded crush force throughout stroke, no hard stop at stroke limit
Seat architecture	Energy-absorbing leg + foam/elastomer cushion combination [OPEN — detailed design]
Function	Limit peak loads to pelvis, spine and lower extremities during hard landing
Works with	8 landing supports + 16 airbags + 5-point restraints + roll-cage
Status	[TARGET/STUDY] — density curve, stroke profile, fire rating, mass, dynamic performance [OPEN]
Mass	Included in 700 kg seats + restraints capsule line
Verification	Dynamic seat tests per FAR 27.562 or CS-29 equivalent; combined drop/sled cases

11. RECOVERY ARCHITECTURE — 3 PRIMARY + 2 RESERVE CANOPIES

Rev 19's 3-primary + 2-reserve architecture is retained. Three large primary canopies are precededented by crewed spacecraft recovery systems at comparable mass. Fewer larger canopies reduce simultaneous-inflation complexity and inter-canopy collision risk at low altitude — the failure mode most common in real rotorcraft emergencies.

Recovery Mode	Use Case	Status
Controlled flight recovery	Recoverable disturbance	[REQ/STUDY]
Autorotation	Engine/power loss when altitude/rotor state permit	[REQ/STUDY]
Selective mass/hazard separation	Fire, structural or combined emergency	[REQ/CONCEPT]
Capsule separation	Catastrophic state	[REQ/CONCEPT]
3 primary recovery canopies	Primary post-separation vertical recovery	[STUDY]
2 reserve canopies	On detected primary failure only	[STUDY]
8 energy-absorbing landing supports	Final ground impact attenuation	[REQ/TARGET]
Crashworthy seats (≥150 mm) + 16 airbags	Occupant load management after ground contact	[TARGET/STUDY]
Post-landing survival	Locator, comms, lighting, emergency power, rescue interface	[REQ]

For 7,000 kg capsule at 7.5 m/s target descent rate: **A_{total} ≈ 2,983 m²**; per canopy (÷3): ≈ **994 m²** each; corresponding diameter ≈ **35–36 m** per primary canopy. Consistent with spacecraft precedent. Detailed sizing: Document 159. The minimum deployment altitude is **[OPEN — CRITICAL]** (OI-02). Most rotorcraft accidents occur at low altitude.

12. CORE SAFETY CASE — REBUILT WITHOUT MAIN-ROTOR SEPARATION

The single largest credibility risk in Rev 12 was the implicit dependency of the safety case on main-rotor separation — a capability for which no credible 6-DOF simulation, structural trajectory analysis, or physical evidence exists. A 26 m rotor carries several hundred megajoules of rotational kinetic energy and a gyroscopic moment resisting rapid attitude change. Removing this dependency from the core safety case is what makes the remaining architecture defensible.

The survival chain works assuming the main rotor is never separated.

- Controlled flight recovery / autorotation — no separation required.
- Selective non-rotor hazard separation — engines, fuel tanks (post-isolation), tail rotor/drive, fire-load panels.
- Capsule release from airframe — capsule departs beneath/behind the still-attached rotor disk.
- Capsule descent under 3+2 canopy system.
- Impact attenuation via eight fixed supports + crashworthy seats + 16 airbags.
- Post-landing survival — independent power, life support, locator, communications.

Main-rotor separation is retained as a labeled future research item (Document 080), explicitly excluded from the licensable core baseline and from primary IP claims. If independent 6-DOF analysis and mechanism test evidence later demonstrate safe separation, it may be reintroduced through the configuration change process.

13. STAGED SEPARATION TIMELINE

Stage	Action	Target Window	Confidence Basis	Status
S0	Validate emergency state; arm path; isolate fuel/electrical/energy	0.0–0.5 s	Comparable to ejection-seat arming	[TARGET]
S1	Fuel isolation and designated fire-hazard separation	0.5–1.8 s	Comparable to crash fuel-shutoff	[TARGET]
S2	Engine + tail-rotor drive separation (non-main-rotor)	1.8–3.5 s	Comparable to cargo-release, scaled	[TARGET/OPEN]
S3	Capsule-to-airframe primary structure release	3.5–5.0 s	OPEN — CRITICAL: no direct certified analog	[OPEN — CRITICAL]
S4	Capsule clearance verification	5.0–5.8 s	Sensor-response-limited	[TARGET]
S5	Recovery canopy authorization	After S4, gated on clearance	Not time-gated; correctness first	[REQ]

If the architecture cannot safely achieve these windows, targets shall be revised under configuration control — not achieved by reducing verification rigor. Partial and asymmetric separation are first-class emergency states at every stage.

14. EMERGENCY STATE MACHINE

State	Entry Condition	Primary Action	Exit / Next
1 — NORMAL FLIGHT	All systems nominal	Continuous monitoring; locks engaged	Anomaly → 2
2 — ANOMALY DETECTED	Sensor/model flags abnormality	Increase monitoring; begin confirmation	Confirmed → 3; Rejected → 1
3 — CONFIRMATION	Potential condition detected	Independent evidence and validity checks	Confirmed → 4
4 — IMPACT ASSESSMENT	Condition confirmed	Estimate authority, altitude, airspeed, fire/structural risk	Complete → 5
5 — RECOVERY ATTEMPT	Assessment complete	Conventional control, OEI, autorotation	Succeeds → 1; Fails → 6
6 — SURVIVAL AUTHORIZATION	Recovery failed	Safety Kernel evaluates all preconditions	Authorized → 7; Not met → degraded
7 — STAGED SEPARATION	Authorized	Execute designated release set (non-rotor)	Complete → 8
8 — CAPSULE CLEARANCE	Separation executed	Verify clearance; identify residual attachments	Clear → 9; Partial → partial-release
9 — RECOVERY DEPLOYMENT	Clearance verified	Authorize parachute within validated envelope	Deployment confirmed → 10

State	Entry Condition	Primary Action	Exit / Next
10 — DESCENT MANAGEMENT	Canopies deployed	Monitor attitude, rates, canopy state, support readiness	Ground imminent → 11
11 — LANDING ATTENUATION	Ground contact	Supports + seats + airbags all load simultaneously	Landing complete → 12
12 — POST-LANDING SURVIVAL	Stabilized	Locator active; comms active; life support active	Rescue confirmed → 13
13 — AUDIT	Rescue complete	Preserve all event data, sensor states, command history	Data secured — END

15. INDEPENDENCE, FAILURE MODES AND COMMON-CAUSE CONTROL

Failure / Hazard	Severity	Required Design Response
Unintended capsule separation in normal flight	Catastrophic	Mechanical locks + electrical inhibits + command validation + monitoring
Partial separation (incomplete component release)	Critical	State sensing + asymmetric-load analysis + defined degraded recovery
Asymmetric separation	Critical	6-DOF dynamic analysis + CG/inertia update + post-release control assessment
Release mechanism failure	Critical	Independent path + defined safe fallback
Sensor loss or bad data	Hazardous	Validity monitoring + redundant/diverse sensing + safe-state logic
L1 automatic detection failure	Hazardous	Independent L2 backup + L3 manual crew initiation
Power-bus failure	Critical	Independent survival power for critical functions
AI/ML model failure	Hazardous	Deterministic Safety Kernel remains authoritative
Fire during separation	Critical	Fuel isolation + fire containment + coordinated sequencing
Released component collision with capsule/parachutes	Catastrophic	Trajectory analysis + clearance envelopes + controlled sequencing
Parachute deployment outside envelope	Critical	Hard inhibits + validated deployment conditions
Landing-support failure (one or more)	Hazardous	8-support redundancy + asymmetric/failed-leg structural analysis
Airbag inadvertent deployment in flight	Critical	Deterministic inhibit logic + dual-path sensing + mechanical safing
Airbag non-deployment on impact	Hazardous	Backup initiation path + sled test verification
Airbag overpressure / gas toxicity in sealed capsule	Critical	OI-18: sealed-volume gas analysis required (Rev 20)
Seat stroke bottom-out at extreme load	Hazardous	Progressive absorber + support stroke + worst-case velocity analysis
Common-cause EMI/lightning / fire	Critical	CCA + physical/electrical segregation + zone separation

16. PRINCIPAL SYSTEM INTERFACES

ID	Definition	Normal-State	Emergency Function	ICD
CAP-AF	Capsule ↔ airframe structural attachment	Transmit normal flight loads	Controlled release on Safety Kernel authorization	Doc 136
CAP-PAR	Capsule ↔ parachute system	Parachutes stowed; deployment inhibited	Deploy primary canopies after clearance verified	Doc 164
CAP-LG	Capsule ↔ eight landing supports	Ground loads during normal operations	Primary crash/landing load path after separation	Doc 156
CAP-AB	Capsule ↔ 16 airbags + sensors/inflators	Stowed; armed only under deterministic preconditions	Deploy under Safety Kernel authorization at impact	Doc 145
CAP-SEAT	Capsule floor ↔ 8 crashworthy seats + restraints	Normal seating load path	Stroke + energy absorption during impact	Doc 123
JET-PWR	Independent release power bus	Standby / trickle charge	Power to all release mechanisms on command	Doc 407
JET-CMD	Automatic/backup/manual command paths	Inhibited; monitoring only	Transmit release commands through independent paths	Doc 406
JET-SNS	Release-state sensing	Report attached state	Confirm released state at each stage	Doc 408
FUEL-ISO	Fuel isolation ↔ separation coordination	Normal fuel flow	Isolate fuel before or concurrent with separation	Doc 104
REC-LOC	Post-landing locator/communications	Dormant / periodic beacon check	Activate on post-separation; transmit position	Doc 405
AV-CTRL	Avionics/control ↔ Safety Kernel	Pass flight data for monitoring	Safety Kernel issues authorization; avionics executes	Doc 241

17. ENGINEERING / IP VALUE SURFACES

Rev 20 concentrates on four primary claim candidates (one added relative to Rev 18/19) and retains remaining elements as supporting system narrative. Four narrowly-scoped, well-supported claims are stronger than nine broad assertions.

IP-A — Deterministic Safety Kernel Gating AI-Assisted Emergency Prediction. The specific combination of: (a) AI/ML-based anomaly prediction; (b) a hard deterministic precondition layer; and (c) selective, state-dependent (not all-or-nothing) actuation in a crewed rotorcraft survival system. Assessed as the strongest single claim in the package.

IP-B — Eight-Point Fixed Capsule-Integrated Landing Support with Asymmetric Failure State Management. The combination of: (a) eight-point, capsule-integrated, fixed, energy-absorbing mechanical support configuration; (b) individual support failure-state monitoring; and (c) state management adapting stability assessment to asymmetric failure configurations.

IP-C — State-Dependent Non-Rotor Selective Separation Sequencing for an Occupied Survival Capsule. Staged, state-dependent sequencing logic for selective release of non-rotor hazards from an occupied survival capsule, with explicit exclusion of the main rotor assembly and capsule-centric clearance verification gating recovery deployment.

IP-D — Integrated Multi-Layer Survival Chain for Crewed Rotorcraft (NEW in Rev 20). The specific combination of: (a) Deterministic Safety Kernel (IP-A); (b) state-dependent non-rotor selective separation (IP-C); (c) eight-point capsule-integrated impact attenuation with asymmetric failure management (IP-B); and (d) distributed canopy recovery — as a unified, interdependent system-architecture applied to a crewed rotorcraft survival capsule. System-level claims are harder to design around than component claims. Prior art (Ka-50: rotor separation + ejection seat; Soyuz: capsule + canopies) cannot anticipate the specific four-element combination. IP-D is assessed as the most commercially valuable claim in the package.

Supporting narrative (not standalone patent candidates): six-layer taxonomy; multi-mode recovery; partial/asymmetric-release management; immutable emergency event record; 16-airbag layout; crashworthy seat stroke specification. No novelty, patentability or freedom-to-operate conclusion is asserted

by this baseline.

18. RISK REGISTER

Risk	Severity	Current Status	Priority
Stage S3 (capsule-to-airframe release) has no direct certified engineering analog	High	Open research item	P0 Critical
Low-altitude canopy deployment envelope may exclude most real accident profiles	High	Open — closure study required	P0 Critical
7,000 kg capsule mass not yet bottom-up verified	Medium	Planning allocation only	P0 Critical
No 6-DOF simulation evidence for any separation stage	High	Gate G4 not started	P0 Critical
Capsule clearance from rotor disk after separation not analyzed	High	Not yet addressed	P0 Critical
16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed	Critical	OI-18 — Rev 20 addition	P0 Critical
Airbag system mass 200 kg planning — detailed design may exceed budget	Medium	Rev 20 corrected estimate	P0 High
Crashworthy seat design — stroke, load-distribution, combined-load cases not closed	Medium	[TARGET/STUDY]	P0 High
Six-layer independence against common-cause not demonstrated	High	CCA not performed	P0 Critical
No professional prior-art search for IP-A, IP-B, IP-C, IP-D	Medium	Prior-art posture only	P1 High
Certification basis not formally engaged with any regulator	Medium	Conceptual reference	P1 High
Main-rotor separation (deferred) remains structurally unanalyzed	Medium	Deferred research track	P2 Medium

19. PRIOR ART POSTURE

Class 1 — Ejection-seat rotorcraft (Ka-50/Ka-52): Rotor-blade-severing + crew ejection is directly relevant prior art for any main-rotor-related separation claim. This is why Rev 20 excludes main-rotor separation from all primary IP claims (IP-A through IP-D).

Class 2 — Crewed spacecraft return capsules: Multi-canopy recovery (Soyuz 3 mains, Crew Dragon 3 mains), crashworthy seating, and independent post-landing life support are prior art for capsule-recovery and life-support elements. IP claims are narrowed to the rotorcraft-specific combination, not the general capsule concept.

Class 3 — Helicopter energy-absorbing landing gear: Energy-absorbing skids and struts exist in certified rotorcraft. IP-B novelty rests on the eight-point capsule-integrated configuration and individual support failure-state management — not on energy absorption in general.

Class 4 — Autonomous vehicle safety interlocks (IEC 61508, ISO 26262, DO-178C): Deterministic safety interlocks are well-developed. IP-A novelty is not in the concept of a safety interlock but in the specific combination of AI prediction + deterministic kernel + selective capsule-centric actuation in a crewed rotorcraft survival system.

Class 5 — Automotive and aviation airbag systems: Airbag systems are prior art. Any IP related to airbags would rest on the capsule-specific integration with the Safety Kernel, the sealed-volume environment, and the combined-attenuation architecture.

Class 6 — IP-D system-level combination: No single prior-art reference combines all four elements of IP-D (Safety Kernel + selective non-rotor separation + eight-point capsule attenuation + distributed canopy recovery) simultaneously in a crewed rotorcraft architecture. This is the basis for IP-D's strength. *A professional prior-art search against IP-A through IP-D is required before any patent filing or broad non-NDA disclosure.*

20. CRITICAL OPEN ITEMS REGISTER

ID	Open Item	Status	Pri	Closure Path
OI-01	Stage S3 capsule-to-airframe release has no certified engineering analog	OPEN — CRITICAL	P0	Mechanism design + 6-DOF + ground demo
OI-02	Minimum reliable canopy deployment altitude for 7,000 kg capsule not closed	OPEN — CRITICAL	P0	Sizing + inflation dynamics + envelope study
OI-03	7,000 kg capsule mass remains a planning allocation	OPEN — CRITICAL	P0	Component mass database + weighing plan
OI-04	No 6-DOF simulation for any separation stage	OPEN — CRITICAL	P0	Model hierarchy + correlation plan (G4)
OI-05	Partial and asymmetric separation detection and degraded recovery not detailed	OPEN — CRITICAL	P0	State sensing + FMEA + test cases
OI-06	Independence of six safety layers against common-cause not demonstrated	OPEN — CRITICAL	P0	CCA + fault injection + power/sensor diversity
OI-07	Capsule clearance from rotor disk after separation not analyzed	OPEN — CRITICAL	P0	6-DOF clearance analysis
OI-08	Occupant injury criteria and seat/restraint dynamic performance not closed	OPEN	P0	Dynamic seat tests + criteria selection
OI-09	Eight-support asymmetric failure and terrain interaction cases not closed	OPEN	P0	FEA + drop test matrix
OI-10	26,000 kg vehicle mass lacks component-level uncertainty tracking	OPEN	P1	Mass control plan + reserve policy
OI-11	Rotor/propulsion power-available vs power-required (OEI) not closed	OPEN	P1	Performance model + engine data
OI-12	Fuel isolation timing coordinated with separation not closed	OPEN	P1	Hazard analysis + sequence timing
OI-13	Professional prior-art search for IP-A through IP-D not performed	OPEN	P1	External IP counsel
OI-14	Certification basis not formally engaged	OPEN	P1	Regulator engagement plan

ID	Open Item	Status	Pri	Closure Path
OI-15	ICDs for CAP-AF, CAP-PAR, CAP-LG, CAP-AB, CAP-SEAT, JET-* not issued	OPEN	P1	ICD set under configuration control
OI-16	16-airbag packaging, placement, mass, inflators, deployment logic not closed	OPEN	P0	Design + sled/impact tests
OI-17	Crashworthy seat stroke and load design (≥ 150 mm, $\leq 6,670$ N) not closed	OPEN	P0	Seat design + dynamic tests
OI-18	16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed	OPEN — CRITICAL	P0	Sealed-volume gas analysis + toxicity + venting/staging (Rev 20)

21. PRELIMINARY PERFORMANCE ANALYSIS

21.1 Hover Power Study Calculation

Using actuator disk theory: $P_{ideal} = T^{3/2} / \sqrt{2\rho A}$ where $T = 26,000 \times 9.81 = 255,060 \text{ N}$; $\rho = 1.225 \text{ kg/m}^3$ (ISA sea level); $A = 530.93 \text{ m}^2$. **$P_{ideal} \approx 3.57 \text{ MW}$ [CALC]** — Study value; FM = 0.75 gives **$P_{shaft} \approx 4.76 \text{ MW}$ [CALC]**. Installed engine power must exceed P_{shaft} to account for profile power, transmission losses (~4–8%), tail rotor power (~8–12% of main rotor), accessory extraction, OEI margins, and hot/high performance. A plausible total installed power requirement is in the range of **6–8 MW**, corresponding to two turboshaft engines of approximately **3,000–4,000 kW each** — consistent with this mass class.

21.2 Mission Profile — Rev 20

Mission Parameter	Target Value	Status	Note
Design range	1,000 km	[TARGET]	Must be closed by mission fuel model
Crew + occupants	2 crew + 8 passengers	[TARGET]	8-person capsule limit
Cruise speed	TBD	[OPEN]	Depends on rotor/drag/engine selection
Cruise altitude	TBD	[OPEN]	Optimized for range/power balance
Takeoff fuel	TBD	[OPEN]	Included in 3,400 kg fuel budget
Cruise fuel	TBD	[OPEN]	Dominant range-driver
Reserve policy	TBD	[OPEN]	Per applicable civil/military requirement
IFR diversion reserve	TBD	[OPEN]	Typically 20–30 min additional fuel
Unusable fuel	~120 kg	[PLANNING]	Included in fuel budget
OEI diversion case	Required	[REQ]	Safety case requirement
Hot/high performance	Required	[REQ]	ISA +20°C at 3,000 m minimum target

The 1,000 km range at 26,000 kg MTOW with two turboshaft-class engines is achievable within the 3,400 kg fuel allocation if cruise SFC is in the range of 0.25–0.35 kg/(kW·h) — consistent with modern turboshaft-class engines — but requires closure of aerodynamic drag, rotor performance, and engine selection. No range claim is made until this closure is complete.

22. TECHNOLOGY READINESS LEVELS — REV 20

New section. TRL assessment per NASA/ESA TRL scale (1–9). Values are engineering judgments pending formal TRL review.

System / Subsystem	TRL Today	Basis	Target G2	Target G8
Six-layer survival architecture concept	TRL 2	Architecture defined; no hardware	TRL 4	TRL 8
Deterministic Safety Kernel (IP-A)	TRL 2	Precondition set defined; no software	TRL 4	TRL 8
Capsule CFRP roll-cage concept	TRL 2–3	Similar CFRP capsule structures exist in spacecraft	TRL 5	TRL 8
Eight fixed landing supports	TRL 2	Similar energy-absorbers exist; capsule-integration novel	TRL 4	TRL 7
Stage S0–S2 (arming, fuel isolation)	TRL 3	Analogous systems exist in military jettison hardware	TRL 5	TRL 8
Stage S3 (capsule-to-airframe release)	TRL 1	No direct analog identified	TRL 3	TRL 7
3+2 canopy recovery	TRL 3	Three-main canopy architecture proven in spacecraft	TRL 5	TRL 7
Crashworthy seats (≥150 mm stroke)	TRL 3–4	Analogous seats exist in military helicopters	TRL 5	TRL 7
16-airbag capsule architecture	TRL 1–2	Airbag technology mature; capsule-integration novel	TRL 3	TRL 6
Sealed-capsule airbag gas management (OI-18)	TRL 1	No analogous system identified	TRL 3	TRL 6
Mass/CG/inertia model	TRL 2	Planning model only	TRL 5	TRL 8
6-DOF separation dynamics model	TRL 1	No simulation started	TRL 4	TRL 8
IP-A through IP-D patent claims	TRL 1	Architecture defined; no legal filing	Prior art searched	Filed

23. COMPETITIVE DIFFERENTIATION

New section. Descriptive comparison, not a claim of superior certified safety performance. All HSH advantages are architectural concepts pending engineering verification.

Feature	Bell 412	AW139	S-92	Mi-26	HSH Rev20
Occupied survival capsule as final landing boundary	X	X	X	X	✓ [CONCEPT]
Deterministic Safety Kernel (AI-gated)	X	X	X	X	✓ [CONCEPT]
State-dependent selective non-rotor separation	X	X	X	X	✓ [CONCEPT]
Eight-point capsule-integrated impact attenuation	X	X	X	X	✓ [CONCEPT]
Distributed canopy recovery (3+2)	X	X	X	X	✓ [STUDY]
16-airbag occupant protection in capsule	X	X	X	X	✓ [TARGET/STUDY]
Six independently-capable survival layers	X	X	X	X	✓ [CONCEPT]
Independent post-separation life support	X	X	X	X	✓ [TARGET]
Explicit asymmetric/partial-failure state management	X	X	X	X	✓ [CONCEPT]
Crashworthy seats (≥150 mm stroke)	Some	Some	Some	Military	✓ [TARGET/STUDY]

Commercial positioning: HSH is not positioned as a replacement for existing helicopter models on cabin space or payload-fraction grounds. It is positioned as a distinct product category — a survivability-first rotorcraft — for operators whose primary requirement is occupant survival in catastrophic emergencies: offshore energy, search-and-rescue, medical evacuation, governmental/military VIP transport.

24. PRELIMINARY CAPSULE GEOMETRY ESTIMATE

New section. Shows geometric plausibility of 8 occupants + airbags + crashworthy seats in a capsule of 7,000 kg. All values pending Document 112 closure.

Geometry Parameter	Preliminary Estimate	Status
Capsule outer diameter	≈ 3.2–3.6 m	[STUDY]
Capsule internal usable diameter	≈ 2.6–3.0 m (after roll-cage wall thickness)	[STUDY]
Capsule internal height (usable)	≈ 1.75–2.0 m (seated + headroom)	[STUDY]
Capsule outer shell thickness (CFRP + fairing)	≈ 100–150 mm per side	[STUDY]
Roll-cage member cross-section	≈ 60–100 mm diameter CFRP tubes	[STUDY]
Seat arrangement	2 rows × 4 seats, or circular/staggered 8-seat layout	[OPEN]
Seat pitch	≈ 650–750 mm	[STUDY]
Seat width	≈ 450–500 mm per position	[STUDY]
Seat stroke height occupied by absorber	≈ 150–200 mm below seat pan	[STUDY]

A circular capsule of ~3.4 m outer diameter, ~2.8 m internal diameter, ~1.85 m internal height accommodates 8 crashworthy seats in a 2×4 configuration with ≈700 mm pitch; 16 airbags stowed in headrests, sill-mounted and seat-back positions; 5-point restraints; emergency exits at minimum 2 locations with airbag-compatible egress paths; and internal volume for life support, emergency power, avionics and fire suppression. Geometric plausibility is supported at this preliminary level — bottom-up closure required (Documents 112, 113).

25. DETAILED AIRCRAFT MASS BREAKDOWN (SUMMARY)

Main Airframe 5,000 kg (fuselage primary 1,450; frames/bulkheads 650; tail boom 520; engine/transmission support 380; rotor mast support 260; skins/fairings 310; doors/panels 190; firewalls 170; fittings 260; cabin floor outside capsule 210; integration margin remainder). Main Rotor 2,100 kg (blades 980; hub 280; mast 260; swashplate 170; pitch links 110; vibration hardware 80; lubrication 40; allowance remainder). Tail Rotor 480 kg (blades 145; hub 65; gearbox 95; drive 60; controls 35; interface remainder). Engines/Transmission 3,100 kg (two engines ~700 each; accessories 180; main gearbox 650; intermediate 180; shafts 100; lubrication 80; mounts 60; exhaust 50; margin remainder). Fuel System + Usable Fuel 3,400 kg (usable fuel ~2,100; tanks 280; pumps 90; lines 110; sensors 55; isolation 75; venting 70; unusable 120; margin remainder). Avionics/Electrical/Hydraulic/Thermal 1,600 kg as allocated in \$MASS ACCOUNTING. All values ASSUMPTION/STUDY pending bottom-up closure.

26. VERIFICATION, VALIDATION AND EVIDENCE PROGRAM

Gate	Objective	Key Evidence Items	Status
G0	Architecture Freeze	Safety philosophy; boundaries; interfaces; emergency states; airbag and seat concepts locked	Approaching closure
G1	Analytical Closure	Mass, CG, inertia, rotor, power, fuel, structural paths, separation kinematics, seat stroke analysis	Not started
G2	Subsystem Demonstration	Release mechanisms, locks, absorbers, parachute, airbag module, seat stroke, sensors, independent power	Not started
G3	Integrated Ground Demonstration	End-to-end sequence; timing; state sensing; capsule clearance; airbag deployment in representative volume	Not started
G4	6-DOF Simulation	Vehicle and capsule dynamics; partial/asymmetric release; parachute inflation; sealed-capsule airbag gas	Not started
G5	Structural / Crash Testing	Capsule, attachments, seats (dynamic stroke per FAR 27.562 proxy), airbag sled tests, supports, energy absorbers	Not started
G6	Recovery System Testing	Canopy deployment; inflation loads; failure cases; stability; landing performance	Not started
G7	Flight Development	Progressive envelope expansion; normal and emergency recovery validated in flight	Not started
G8	Certification / Qualification	Full evidence package against selected certification basis	Not started

As of Rev 20: all HSH-00 claims are Specified or Studied. None are Analyzed, Simulated, Tested, Verified, Validated, Qualified, or Certified. Terminology: Specified = defined in design; Analyzed = documented analytical result; Simulated = supported by a model; Tested = demonstrated in a defined test; Verified = evidence shows compliance with a requirement; Validated = suitability established for intended use; Qualified/Certified = formal processes complete.

27. ENGINEERING MATURITY FRAMEWORK

Configuration Item	Rev 20 Control Rule	Closure Evidence Required
MTOW / design mass	Freeze only after bottom-up mass model and reserve policy reconciled	Mass report + uncertainty analysis
Rotor geometry	Maintain 26.0 m target until performance study closes	Rotor sizing + power model
Capsule mass	7,000 kg planning target; no closure claim	Bottom-up mass budget (Doc 113)
Occupancy	8 persons maximum — LOCKED for Generation 1	Configuration control only
Recovery	3 primary + 2 reserve study architecture	Sizing + inflation dynamics + failure testing
Separation	Non-rotor candidate set only in core baseline	6-DOF + clearance + ground demonstration
Safety authorization	Deterministic kernel remains authoritative	Requirements + software/hardware verification
Airbags	16 total (2 per occupant) TARGET/STUDY	Packaging + sled/impact tests + OI-18 gas analysis
Crashworthy seat stroke	≥150 mm stroke; ≤6,670 N criterion	Dynamic seat tests + analysis

28. MASTER CONFIGURATION RECORD

Configuration Item	Rev 20 Baseline	Status
Aircraft programme name	HSH Heavy Safe Helicopter	LOCKED
Founder	Amirbahador Atai	CONTROLLED
Generation	Generation 1	LOCKED
Document number	HSH-00	LOCKED
Revision	Rev 20	BASELINE
Capsule occupancy maximum	8 persons	LOCKED LIMIT
Vehicle design mass	26,000 kg	TARGET / OPEN
Capsule mass allocation	7,000 kg	PLANNING / OPEN
Main rotor target	26.0 m	TARGET / OPEN
Tail rotor target	4.0 m	TARGET / OPEN
Range target	1,000 km	TARGET / OPEN
Landing supports	8 fixed energy-absorbing	CONCEPT / OPEN
Recovery	3 primary + 2 reserve canopies	STUDY / OPEN
Occupant airbags	16 (2 per occupant)	TARGET / STUDY
Crashworthy seat stroke	≥150 mm; ≤6,670 N	TARGET / STUDY
Core separation	Selective, staged, non-rotor only	CONCEPT / OPEN
Main-rotor separation	Excluded from core safety case	DEFERRED RESEARCH
Safety authorization	Deterministic Safety Kernel	CONCEPT / OPEN
IP claims	IP-A, IP-B, IP-C, IP-D	CONCEPTUAL / NOT FILED
Development gate	G0 approaching closure	CURRENT

29. ENGINEERING BASIS AND EXTERNAL REFERENCES

Predecessor documents: LSH-00 Rev3.2 (six-layer philosophy; independent initiation; controlled separation; eight supports; capsule-as-final-boundary concept); USCC Doc 00 Rev1.3 (jettison sequence; deterministic safety authorization); HSC-00/HSH-00 Rev 12–19 (progressive engineering development and correction).

External engineering references: EASA CS-29 Amendment 12 (March 2026): Large rotorcraft certification standard; jettisonable configurations require clear trajectory, no damaging interaction, acceptable post-jettison stability and performance. FAR 27.562 / CS-27 Dynamic seat test requirements: Proxy reference criterion for ≤6,670 N peak spinal compressive load used in crashworthy seat specification (§10B). MIL-STD-1290A: Helicopter crashworthiness design criteria; relevant to capsule structural design, seat design, landing-support design and occupant injury criteria. NASA TRL scale: Used for TRL assessment in §22.

30. DESIGN PHILOSOPHY — CONTROLLED SURVIVAL, NOT A SINGLE DEVICE

HSH Generation 1 integrates multiple survival mechanisms into one state-aware architecture. The vehicle does not depend solely on a parachute, solely on autorotation, solely on a detachable cabin, or solely on automatic software. Removing main-rotor separation from the core safety case — introduced in Rev 13 and retained through Rev 20 — **strengthens** rather than weakens the layered proposition: a safety case that requires its riskiest, least-proven capability to succeed is not actually layered. Rev 20 extends the impact-attenuation layer with 16 airbags and crashworthy seats specified to the correct engineering standard — these are complementary protections, not substitutes for structural crashworthiness, landing supports, or recovery canopies. The strongest future version of HSH Generation 1 is the version in which every important claim has a visible path from requirement to model to test to controlled evidence. Documentation is not a substitute for evidence.

31. FINAL GENERATION 1 BASELINE STATEMENT

HSH-00 Rev 20 is the definitive Generation 1 conceptual master baseline for the HSH Heavy Safe Helicopter. It is the document a technical due-diligence team from a potential IP licensee should receive as the starting point for evaluation.

What is defined: the full six-layer survival architecture; the Safety Kernel and its precondition set; the staged S0–S5 separation timeline with honest confidence-basis assessment; the 3+2 canopy recovery architecture with preliminary sizing; the 8-point capsule-integrated landing-support system; the 16-airbag and crashworthy-seat occupant-protection architecture (with corrected engineering specifications); four IP claims (IP-A through IP-D) with prior-art awareness; a full open-items register (OI-01 through OI-18); a TRL assessment; a competitive differentiation table; a preliminary capsule geometry check; and a mission profile framework.

What remains to be proven: everything in the open-items register. The gap between architecture and evidence is the scope of the Generation 1 engineering document tree (Documents 001–500).

END OF HSH-00 REVISION 20

HSH Heavy Safe Helicopter | Generation 1 | Founder: Amirbahador Atai
Conceptual / IP Baseline | Proprietary Concept Development Material
Not a certification approval | Not a filed patent | Not demonstrated performance